

5. TEST ITERATIVELY

Testing is an opportunity to learn about your solution and your user.

What?

The Test mode is when you solicit feedback, about the prototypes you've created, from your users and have another opportunity to gain empathy for the people you're designing for. Testing is another opportunity to understand your user, but unlike your initial empathy mode, you've now likely done more framing of the problem and created prototypes to test. A rule of thumb: always prototype as if you know you're right, but test as if you know you're wrong – testing is the chance to refine your solutions and make them better.

Why?

Testing is made essential with the need to refine prototypes and solutions, to learn more about your user and to refine your POV by revealing insights such as that not only did you

**PROTOTYPING IS
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MANAGE THE SOLUTION
BUILDING PROCESS.**

to compare between the various prototypes and pick one that suits his needs.

MAKE THE PROCESS YOUR OWN BY ITERATION

Iteration is a fundamental of good design. Iterate both, by cycling through the process

lenges can be taken on by using the design modes in various orders; furthermore, there are an unlimited number of design frameworks with which to work. The process presented here is one suggestion of a framework. Ultimately you'll make the process your own and adapt it to your style and work. Hone your own process that works for you. Most importantly, as you continue to practice innovation you take on a designers' mindset that permeates the way you work, regardless of what process you use.

FROM A BUSINESS POINT OF VIEW ADVENTURES IN DESIGN THINKING

- Roger Chandler, Director of Business Planning & Innovation, Intel Corporation.

One of the more exciting things happening at Intel® these days is our increased focus on user experiences. A shift like this can require new ways of thinking, innovating, and collaborating – particularly in SSG (Software and Services Group), since software plays a major role in translating platform features into meaningful experiences. One user-focused innovation model our team has started utilizing is "design thinking". Similar to methodologies or mental models such as Agile, Decision Quality and Lean Start-up, design thinking is an iterative process to help teams focus, solve problems, and innovatively move forward. It's relatively easy to implement, can help a team establish a strong user focus and can improve collaboration amongst teams inside and outside of Intel®.

I was first introduced to the concept of "design thinking" eleven years ago, when I was working on a project in the Intel® Labs. Our team's challenge was pretty broad in scope – define a vision for how the proliferation of wireless technologies could improve the lives of billions of people around the world. As part of this project, our team partnered with the legendary innovation firm, IDEO, to identify user needs that future Intel® technologies could address. The joint team created mock-ups of context aware mobile devices, storyboarded networks of wireless sensors that could increase global agricultural efficiency, and created "low-resolution" prototypes of rugged, connected, bio-sensor enabled devices that could save lives in hard-to-reach rural areas. Several of the use cases from the project influenced technolo-



CyberLink MediaStory streamlines the process of organising photos and videos making it easy to share personal stories with touch support for Windows 8

not get the solution right, but you also failed to frame the problem correctly.

How?

To test effectively, show the user the prototype and don't explain everything to him – watch how he interacts with it and then listen to the question he has. Your prototypes should be tested in a way that feels like an experience as opposed to an explanation of the solution put forward for evaluation. Finally, ask the user

multiple times, and by iterating within a step – for example, by creating multiple prototypes or trying variations of a brainstorming topic with multiple groups. Generally, as you take multiple cycles through the design process your scope narrows and you move from working on the broad concept to the nuanced details, but the process still supports this development.

For simplicity, the process is articulated here as a linear progression, but design chal-